Aniket Bembale

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aniket-bembale

Summary

AI Engineer with 2.8+ years of experience building and deploying GenAI and ML solutions. Proficient in Python, AWS, RAG, LLMs, A2A, MCP and MLOps, with strong expertise in data pipelines, model development, and scalable system design. Experienced in CI/CD, performance optimization, and rapid POC development.

SKILLS

Programming Languages: Python, SQL, C, C++, Java
AI/ML & GenAI: Multi-Agent Systems, Machine Learning, Deep Learning, NLP, Generative AI, Advanced RAG, LLM Fine-Tuning
Cloud & DevOps: AWS (Lambda, Bedrock, EC2, S3, ECR), Docker, CI/CD, MLflow
Tools & Platforms: Git, GitHub, Streamlit, Flask, FastAPI, Claude, CrewAI, AutoGen,LlamaIndex, Langchain, Langgraph, Kiro
Databases & Vector Stores: MySQL, PostgreSQL, MongoDB, FAISS, ChromaDB, Qdrant,OpenSearch, S3 Vectors

Experience

Senior Research Associate (AI/Data) 2026 - Present
CloudThat, Bangalore, India

- Developed a GenAI conversational analytics platform using **AWS Bedrock, Lambda, and DynamoDB, supporting intelligent tool routing, parallel API calls**, and multi-turn analytics across **400+** cinema locations.
- Built an AI customer support chatbot on AWS using using a Strands Agent framework orchestrated by Amazon Nova Pro (Bedrock) with a hybrid **RAG + Text-to-SQL** architecture for product/FAQ and live order/account queries.
- Reduced LLM token usage by **40%** and per-query operational cost through optimized tool-selection and prompt design on Amazon Bedrock.

Project Associate (AI/ML Engineer) Feb 2024 - Present
C-DAC CINE, IIT Guwahati, India

- Developed a GenAI-based predictive policing system using **RAG** using Llama Model, improving insight accuracy and reducing manual analysis effort by **40-50**% through automated charge-sheet generation.
- Engineered a real-time evacuation optimization pipeline combining **XGBoost-based dynamic edge weighting** with **Dijkstra's algorithm**.
- Built a **hybrid quantum–classical neural network (HQNN)** on 6 flood features, achieving performance comparable to classical models using quantum-enhanced gradient updates for real-time evacuation routing.
- Devised a **Quantum Computing approach for Graph Optimization**, focusing on the Shortest Path Problem to explore more efficient than classical methods.
- Collaborated with R&D project under the **National Quantum Mission (MeitY)**, focusing on Quantum Machine Learning algorithms for disaster escape routing

Projects

Intelligent Document-to-JSON Parser using LLM and LangChain

- Built an LLM-based pipeline to convert unstructured documents into structured JSON using GPT-4 + LangChain.
- **Tech**: GPT-4, LangChain, TrOCR, FAISS, FastAPI, Docker

AI-Powered Budget Travel Planner

- Built an agentic AI system for end-to-end itinerary validation and budget booking, using **SerpAPI (flights/hotels) and LangGraph with Google Gemini** for reasoning, and deployed via Streamlit-based frontend.

Publications

- Aniket Bembale, Nagendra Singh, Vinayaka Poorma, Jitesh Choudhary. "***Quantum Computing for Graph Optimization: An Approach to the Shortest Path Problem,***" ETTIS 2025 Conference. **(Best Paper Award)**
- A. Vinil, P. Iyer, J. Chetan, A. Bembale, and N. Singh. "***A Topical Review of Quantum and Classical Machine Learning Approaches to Disaster Escape Routing Problems,***" *IEEE Access*, 2025.
- Vinayaka Poorma, Nagendra Singh, Aniket Bembale, Jitesh Choudhary. ***FloodGraph: A Data Re-Uploading Enabled QSVR Framework for Flood Parameter Prediction and Dynamic Graph Updation*** (Accepted, GCON 2026)

Certifications

AWS Certified Machine Learning Engineer – Associate Aug 2026
AWS Agentic AI Demonstrated Aug 2026
AWS Serverless Demonstrated Aug 2026

Education

Center for Development of Advanced Computing Noida, Uttar Pradesh, India
Post-Graduation Diploma | Artificial Intelligence 2022 - 2023
Savitribai Phule Pune University Pune, Maharashtra, India
Masters Degree | Science 2020 - 2022
Savitribai Phule Pune University Pune, Maharashtra, India
Bachelor's Degree | Science 2017 - 2020